(50 marks) Question 1. A 400V EV battery system is interfaced (via an OBC) to a single-phase 240V-60Hz source that can support up to 10kW of power via phases a and b outputs. The inverter is operated at a switching frequency of F_{sw} = 10kHz. The OBC has also an APM module featuring an auxiliary inductor and capacitor.

a) (15 marks) Design the circuit parameters "L_g", "L_aux" and "C_aux" based on the following criteria:

- The current ripple through "L_g" must not exceed 5% of the rated current.
- The resonant frequency F_r due to "L_aux" and "C_aux" must be 60Hz<F_r<<F_{sw}. (i.e. capacitive reactance must dominate at 60Hz)

Show your calculations and cite any sources.

Design Calculations for Onboard Charger (OBC)

1. *Given Parameters*

- Output power: $P = 10\,\text{kW}$
- Grid voltage (RMS): $V_{\text{RMS}} = 240\text{ V}$
- Power factor: Unity
- Battery voltage: $V_{\text{DC}} = 400\text{ V}$
- Switching frequency: $f_{\text{sw}} = 10\,\text{kHz}$
- Allowed current ripple: $\Delta I = 0.05 \times I_{\text{rated}}$

2. *Rated Current* [1]

$$I_{\text{rated}} = \frac{P}{V_{\text{RMS}}} = \frac{10{,}000}{240} \approx 41.67\text{ A}$$

3. *Maximum Duty Cycle* D [1]

$$D = \frac{V_{\text{peak}}}{V_{\text{DC}}} = \frac{\sqrt{2} \times 240}{400} \approx 0.85$$

4. *Grid Inductor* L_g [1]

$$L_g = \frac{D \cdot V_{\text{DC}}}{f_{\text{sw}} \cdot \Delta I} = \frac{0.85 \cdot 400}{10{,}000 \cdot 2.083} \approx 16.35\text{ mH}$$

(Note: $\Delta I = 0.05 \times 41.67 \approx 2.083$ A)

*5. **Auxiliary Capacitance** C_{aux}* [1]

Given:

- Ripple power $P_{\text{ripple}} = 100\,\text{W}$
- Ripple frequency $f = 120\,\text{Hz}$
- Peak AC voltage:

$$|v_{\text{aux}}| = \sqrt{2} \cdot 240 = 339.4\,\text{V}$$

Formula:

$$C_{\text{aux}} = \frac{2 \cdot P_{\text{ripple}}}{\omega \cdot (v_{\text{aux, peak}})^2} = \frac{2 \cdot 100}{2\pi \cdot 60 \cdot (339.4)^2} \approx \cancel{460\,\mu\text{F}}$$

*6. **Auxiliary Inductance** L_{aux}* [1]

To achieve a resonant frequency of:

$$f_r = 1000\,\text{Hz}$$

Use resonance condition:

$$f_r = \frac{1}{2\pi\sqrt{L_{\text{aux}} C_{\text{aux}}}} \Rightarrow L_{\text{aux}} = \frac{1}{(2\pi f_r)^2 C_{\text{aux}}}$$

Substitute:

$$L_{\text{aux}} = \frac{1}{(2\pi \cdot 1000)^2 \cdot 460 \times 10^{-6}} \approx \cancel{55.1\,\mu\text{H}}$$

Reference

[1] Texas Instruments, *1.6-kW, Bidirectional Micro Inverter Based on GaN Reference Design (Rev. A)*, Design Guide TIDUF63A, Dec. 2023, revised Jun. 2024. [Online]. Available: https://www.ti.com/lit/pdf/TIDUF63

b) (20 marks) Build the grid-connected OBC circuit and the current + APD controllers in Simulink as shown in the attached pdf file. The oscillator blocks consist of two transfer functions (see lecture 14). Set the parameter k for each oscillator as follows:

- Voltage Oscillator: k = 100
- Current Oscillator: k = 350
- Power Oscillator: k = 50 for the α and k = 150 for the β. Put the β-generator in series with the α-generator as hinted through the "Power Oscillator" subsystem.

Show screenshots of your final circuit and control model with all the subsystems.

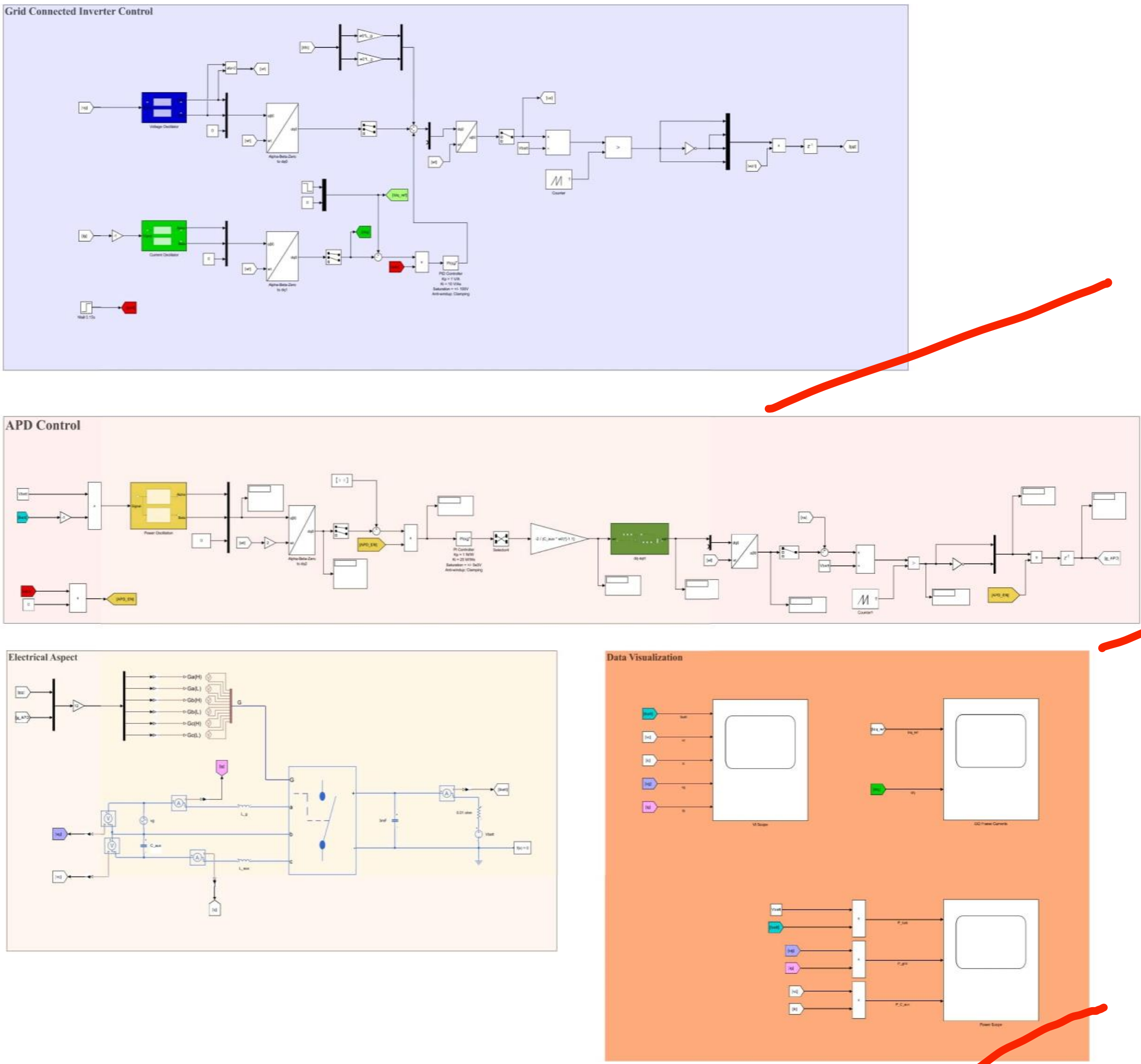


Fig: screenshots of final circuit and control model with all the subsystems.

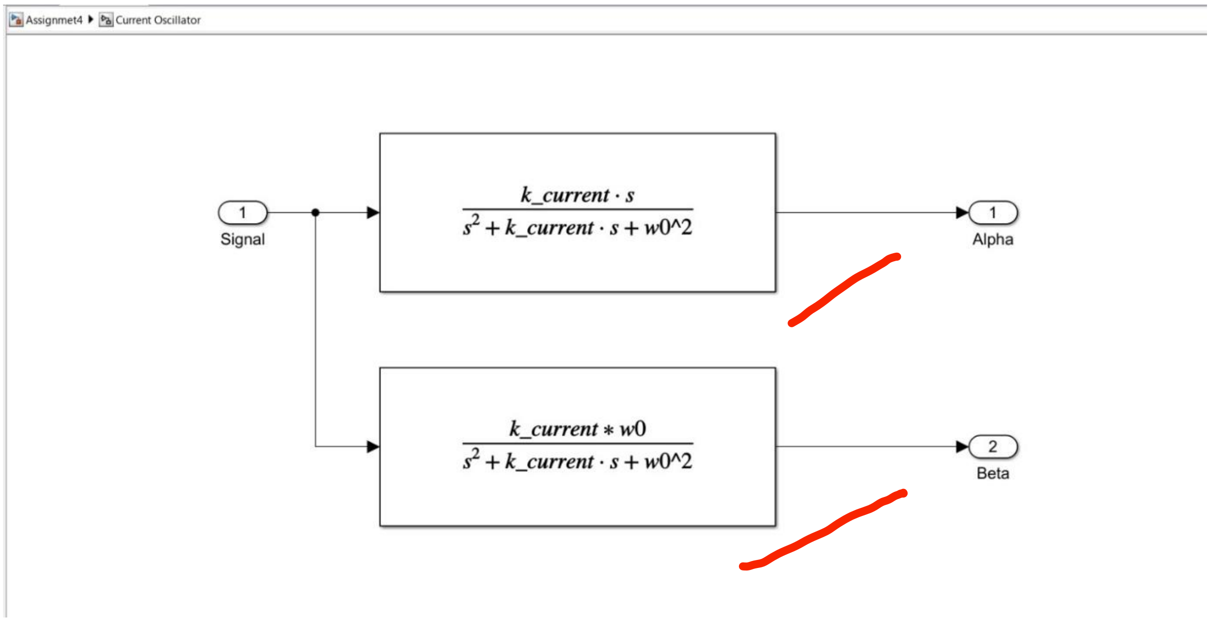


Fig: Current Oscillator

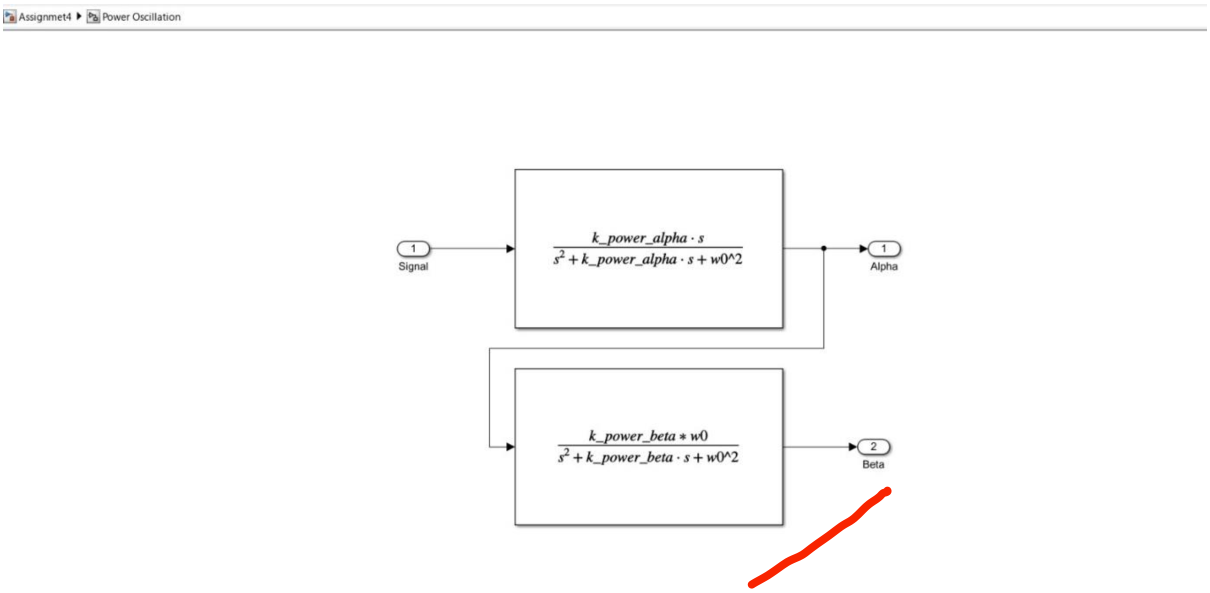
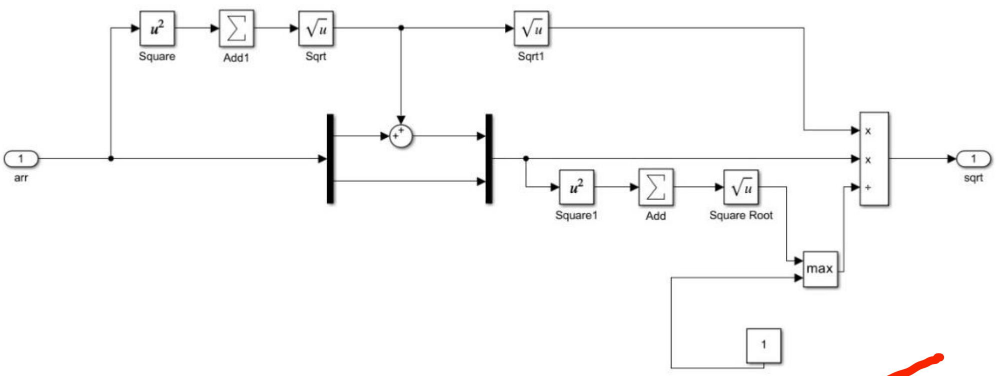


Fig: Power Oscillation



c) (15 marks) Briefly explain why the oscillators $\alpha - \beta$ in the APD controller were connected in series. Would it affect the APD operation if they were in parallel (hint: this question can be correctly answered as yes or no depending on your rationale)?

Note: This question requires you to have a running model.

The alpha (α) and beta (β) signals of voltage and current can be handled separately, or in parallel, because they're independent also you can calculate each one without needing the other. But for the power oscillator used in Active Power Decoupling (APD), it's a different story.

In APD, the α and β signals must be connected in series, because they work together to form a rotating signal called a phasor:

$$\vec{x}(t) = x_\alpha + jx_\beta$$

This phasor is important because it helps detect and cancel out the second harmonic ripple (an unwanted 2ω oscillation) that appears in the power of single-phase systems:

$$\vec{r}(t) = \frac{1}{2}\hat{v}_{ac}\hat{\imath}_{ac}^*$$

$$\Re\{\vec{r}(t)\} = \frac{1}{2}\hat{v}_{ac}\hat{\imath}_{ac}\cos(2\omega t + \theta_i)$$

By combining α and β in series to create this rotating signal, the controller can move it into the dq-frame and use a PI controller to cancel the ripple. The APD system then calculates the voltage it needs to apply using:

$$\vec{v}_{dq,aux} = \sqrt{\frac{-2j}{\omega C_{aux}}\vec{r}_{dq,aux}}$$

If we were to connect α and β in parallel instead, they wouldn't form a proper phasor they'd behave like two unrelated signals. The controller wouldn't be able to process them correctly, and it wouldn't be able to cancel out the ripple. This would cause the capacitor voltage to ripple more and reduce the overall power quality.

So, connecting the α and β power signals in series is necessary to make the APD work properly and keep the power smooth and stable.

(50 marks) Question 2. Configure Simulink to have a fixed time step of 1e-6. Make sure the Simscape solver configuration is set to local solver – Trapezoidal Rule – with simulation time step of 1e-6. Set all axis transformations to “Aligned with phase A axis”. Set a step reference

I_d to the rated current at the 0.2s mark (note: according to the provided example, negative I_d means charging; hence a negative reference is given).

a) (15 marks) With the APD disabled, run the model for 1 second. Show screenshots of the results on the 3 scopes.

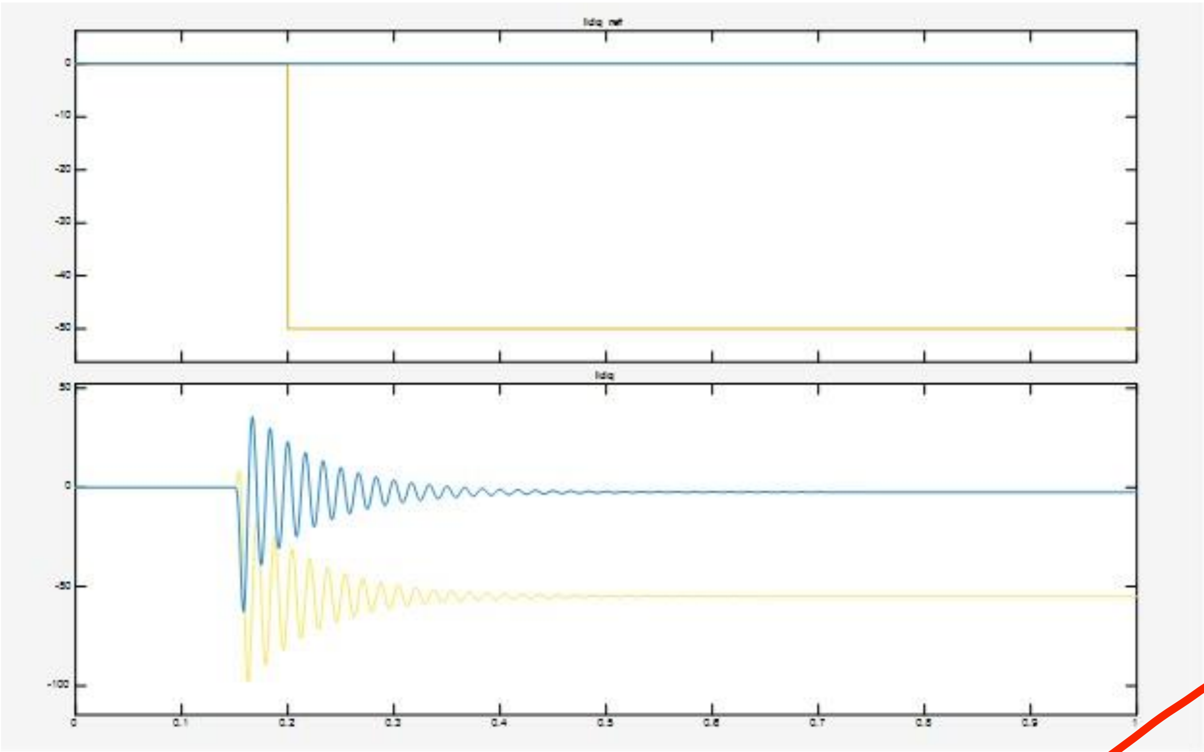
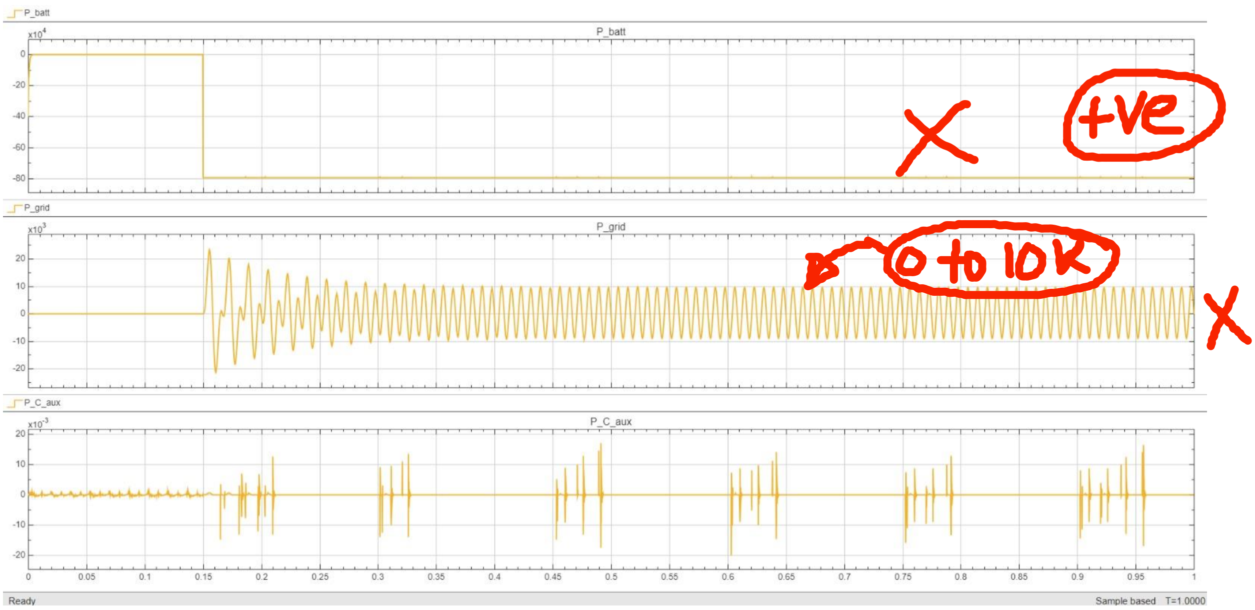


Fig: Idq vs Idq Reference


Ibatt
vc
ic
vg, ig
Ready
Sample based T=1.0000
no ripple?


P_batt
P_grid
P_C_aux
Ready
Sample based T=1.0000
+ve
0 to 10K

```matlab
%% ELEC-8900-58 Assignment 4 – Passive Component Design (Final Version)
clear
clc

%% 1. General Setup
f_grid = 60;                    % Grid frequency [Hz]
w0 = 2 * pi * f_grid;
f_sw = 10000;                   % Switching frequency [Hz]
T_sw = 1 / f_sw;                % Switching period [s]

%% 2. Given Power System Parameters
P_rated = 10000;                % Output power [W]
V_ac_rms = 240;                 % Grid RMS voltage [V]
V_ac_peak = sqrt(2) * V_ac_rms; % Grid peak voltage [V]
Vbatt = 400;                    % Battery voltage [V]
I_rated = P_rated / V_ac_rms;   % Rated RMS current [A]
delta_I = 0.05 * I_rated;       % Allowed current ripple (5%) [A]
Id_ref = -P_rated / Vbatt;      % D-axis reference current [A]
R_batt = 0.01;                  % Battery model resistance [Ohms]

%% 3. Maximum Duty Cycle (Worst-case)
D = V_ac_peak / Vbatt;          % Duty cycle at peak voltage
D_max = D;                      % Use as max duty cycle

%% 4. Grid Inductor Lg (Based on Ripple Constraint)
L_g = (D_max * Vbatt) / (f_sw * delta_I);  % [H]

%% 5. Auxiliary Capacitance C_aux (From design)
C_aux = 460e-6;                 % 460 µF

%% 6. Auxiliary Inductance L_aux (From design)
L_aux = 55.1e-6;                % 55.1 µH

%% 7. Oscillator Transfer Function Parameters
k_voltage = 100;
k_current = 350;
k_power_alpha = 50;
k_power_beta  = 150;

%% 8. PI Controller Parameters (dq current loop) – Tunable
PI_current.Kp = 1;
PI_current.Ki = 10;
PI_current.Sat = 100;

%% 9. Display Summary
fprintf('\n--- Assignment 4 Passive Component Summary ---\n');
fprintf('Grid Inductor Lg         = %.2f mH\n', L_g * 1e3);
fprintf('Auxiliary Inductor Laux  = %.2f uH\n', L_aux * 1e6);
fprintf('APD Capacitor Caux       = %.2f uF\n', C_aux * 1e6);
fprintf('D-axis Current Id_ref    = %.2f A\n', Id_ref);
fprintf('Battery Resistance R     = %.3f Ohm\n', R_batt);
fprintf('Switching Frequency      = %.0f Hz\n', f_sw);
```

b) (15 marks) Enable the APD module and rerun the model for 1 second. Show screenshots of the results on the 3 scopes.

Caux is not handling any power. APD is not working.

c) (20 marks) Discuss the results following the key points:

- Without the APD, how does the battery power look? Is it exactly 10kW?
- After enabling the APD, how does the battery power look? Is it exactly 10kW?
- Compare the 3 measured powers in each case. Did the grid power change when the APD was enabled?

2(c) Effect of APD on Battery Power and Grid Power

Without APD:

When Active Power Decoupling (APD) is disabled, the inverter draws power from the single-phase grid and delivers it directly to the battery. Since single-phase systems cause power to fluctuate at twice the grid frequency (typically 120 Hz), this ripple is passed straight to the battery. As a result, the battery power is not steady and contains oscillations. Even if the average power is close to 10 kW, the waveform shows noticeable variations. This kind of ripple can be harmful to the battery and reduces charging efficiency.

With APD enabled:

When APD is turned on, the auxiliary circuit is responsible for absorbing the 120 Hz ripple. This decouples the fluctuating power from the battery, so the battery receives a much smoother and more stable DC charging power. The battery power waveform becomes nearly constant and better aligned with the desired charging level, typically around 10 kW. This improves the reliability and lifespan of the battery system.

Comparison of Powers:

The power drawn from the grid remains nearly the same in both cases, close to the commanded charging power. What changes is how that power is distributed. Without APD, all the ripple is sent to the battery. With APD, the ripple is filtered out by the APD branch, and the battery sees mostly clean DC power. The total energy from the grid does not increase, but the *quality* of the power reaching the battery significantly improves.